Şirag Erkol

Contact Information

Northwestern University
Center for Science of Science and Innovation
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Academic Positions

Northwestern University, Evanston, IL — **July 2025 – present**
Research Assistant Professor

Northwestern University, Evanston, IL — **July 2023 – June 2025**
Postdoctoral Fellow

Indiana University, Bloomington, IN — **August 2017 – May 2023**
Research & Teaching Assistant

Boğaziçi University, Istanbul, Turkey — **October 2014 – August 2017**
Research & Teaching Assistant

Education

Indiana University, Bloomington, IN — **August 2017 – May 2023**

Ph.D. in Informatics, Complex Networks and Systems track

- Advisor: *Prof. Filippo Radicchi*
- Thesis: *Identifying Influential Spreaders in Complex Networks*
- Minor: Computer Science

M.S. in Informatics

Boğaziçi University, Istanbul, Turkey

M.S. in Industrial Engineering — **September 2014 – August 2016**

- Advisor: *Assoc. Prof. Gönenç Yücel*
- Thesis: *Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics*

B.S. in Industrial Engineering — **September 2010 – June 2014**

- Graduation Project: *Analyzing Parking Stand Requirements for Istanbul Atatürk Airport*

Work Experience

Pacific Northwest National Laboratory, Richland, WA — **June 2021 – August 2021**
Ph.D. Intern

Honors & Awards

IU Luddy Summer Research Award ($4400) — **2022**

Publications

Published:

Crowding controls the scaling of bus frequency with demand
S. Patwardhan, **Ş. Erkol**, F. Radicchi, M. Barthelemy, *PNAS*, 123, e2535998123 (2026)
[DOI] [arXiv]

Symmetry breaking in optimal transport networks
S. Patwardhan, M. Barthelemy, **Ş. Erkol**, S. Fortunato, F. Radicchi, *Nature Communications*, 15, 3758 (2024)
[DOI] [arXiv]

Consistency pays off in science
Ş. Erkol, S. Sikdar, F. Radicchi, and S. Fortunato, *Quantitative Science Studies*, 4, 491 (2023)
[DOI] [arXiv] [GitHub] [Website]

Effective submodularity of influence maximization on temporal networks
Ş. Erkol, D. Mazzilli and F. Radicchi, *Physical Review E*, 106, 034301 (2022)
[DOI] [arXiv]

Who is the best coach of all time? A network-based assessment of the career performance of professional sports coaches
Ş. Erkol and F. Radicchi, *Journal of Complex Networks*, 9, cnab012 (2021)
[DOI] [arXiv] [Website]

Influence maximization on temporal networks
Ş. Erkol, D. Mazzilli and F. Radicchi, *Physical Review E*, 102, 042307 (2020)
[DOI] [arXiv]

Systematic comparison between methods for the detection of influential spreaders in complex networks
Ş. Erkol, C. Castellano and F. Radicchi, *Scientific Reports*, 9, 15095 (2019)
[DOI] [arXiv]

Influence maximization in noisy networks
Ş. Erkol, A. Faqeeh and F. Radicchi, *Europhysics Letters*, 123, 58007 (2018)
[DOI] [arXiv]

Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics
Ş. Erkol and G. Yücel, *International Journal of Modern Physics C*, 28, 1750122 (2017)
[DOI]

Presentations

Talks:

Second acts in scientific and creative careers — **October 2026**
CCS 2026, Binghamton, NY, USA

Second acts in scientific and creative careers — **June 2026**
NetSci 2026, Boston, MA, USA

Effective submodularity of influence maximization on temporal networks — **July 2022**
NetSci 2022, Online

Influence maximization on temporal networks — **July 2021**
Networks 2021, Online

Influence maximization on temporal networks — **March 2021**
University of Limerick, Ireland, Online

Posters:

Second acts in scientific and creative careers — **June 2026**
ICSSI 2026, Boulder, CO, USA

Consistency pays off in science — **June 2023**
ICSSI 2023, Evanston, IL, USA

Systematic comparison between methods for the detection of influential spreaders in complex networks — **September 2020**
NetSci 2020, Online

Influence maximization in noisy networks — **May 2019**
NetSci 2019, Burlington, VT, USA

Community Service

Journal Reviewer: Applied Mathematics and Computation; Chaos, Solitons & Fractals; EPJ Data Science; Expert Systems with Applications; IEEE Transactions on Network Science and Engineering;

Journal of Complex Networks; Journal of Computational Science; Pattern Recognition; Physical Review E; Royal Society Open Science; Science Advances

Conference Reviewer: ISSI (2023); ICSSI (2024); IC2S2 (2026)

Journal Subreviewer: PLOS ONE, PNAS

COMPUTER SKILLS

Frequent user of Python, LaTeX
Experience in Django, NetLogo, HTML, CSS
Familiar with MySQL, C, R, Git, Stella

LANGUAGES

English (fluent), Turkish (native), Armenian (native), French (beginner), Spanish (beginner)

Last updated: October 2026